9.5 The Ratio and Root Tests

(Here, $a_n > 0$)

Given a series $\sum a_n$, the Ratio Test examines the ratio a_{n+1}/a_n to measure its rate of growth. For example, the geometric series $\sum r^n$ has a constant rate of ratio r and it converges if and only if $|r| < 1$.

Theorem (**The Ratio Test**): Let $\sum a_n$ be a series with positive terms and suppose that

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \rho.$$

Then ($\sum a_n$)

(a). the series converges if $\rho < 1$,

(b). the series diverges if $\rho > 1$ or ρ is infinity,

(c). the test is inconclusive if $\rho = 1$.

Proof of the Ratio Test: $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \rho$

(a) $\rho < 1$: Pick r in between ρ & 1 and let $\varepsilon = r - \rho > 0$ [$r < 1$]

By the def^n of limit, there is some N st for all $n \geq N$, we have

$$\left|\frac{a_{n+1}}{a_n} - \rho\right| < \varepsilon = r - \rho$$

$$\Rightarrow a_{n+1} < r\, a_n .$$

So
$$a_{N+1} < r\, a_N$$
$$a_{N+2} < r\, a_{N+1} < r^2 a_N$$
$$a_{N+3} < r\, a_{N+2} < r^3 a_N$$
$$\vdots$$
$$a_{N+m} < r^m a_N$$

the geometric series $\sum_{n=1}^{\infty} r^n a_N$ converges because $|r| < 1$. So by the DCT,

$\sum_{n=N+1}^{\infty} a_n$ converges

$$\Rightarrow \sum_{n=1}^{\infty} a_n \text{ converges.}$$

□.

(b) $\rho > 1$: For some N, we have $\frac{a_{n+1}}{a_n} > 1$ for all $n \geq N$.

So $a_N < a_{N+1} < a_{N+2} < \cdots$.

This strictly increasing sequence shows that the nth term does not approach 0. Hence by the nth term test, the series $\sum a_n$ diverges.

(c) If $\lim \frac{a_{n+1}}{a_n} = 1$, we cannot say anything about $\sum a_n$ using ratio test.

Example: $\sum \frac{1}{n^2}$ converges but $\sum \frac{1}{n}$ diverges

both have $\lim \frac{a_{n+1}}{a_n} = 1$.

Examples:

1. $\sum_{n=0}^{\infty} \underbrace{\frac{2^n+5}{3^n}}_{a_n}$

$$\frac{a_{n+1}}{a_n} = \frac{(2^{n+1}+5)/3^{n+1}}{(2^n+5)/3^n} = \frac{1}{3} \cdot \frac{2+5/2^n}{1+5/2^n} \longrightarrow \frac{2}{3} \quad (<1)$$

So by the ratio test, the series $\sum a_n$ converges.

2. $\sum_{n=1}^{\infty} \underbrace{\frac{(2n)!}{n!n!}}_{a_n}$:

$$\frac{a_{n+1}}{a_n} = \frac{(2n+2)!/(n+1)!(n+1)!}{(2n)!/n!\,n!}$$

$$= \frac{(2n+2)(2n+1)}{(n+1)(n+1)} \longrightarrow 4 \quad (>1)$$

By the ratio test, $\sum \frac{(2n)!}{n!\,n!}$ diverges.

Alternatively, $a_n = \frac{(2n)!}{n!\,n!} = \binom{2n}{n} > 0$ for every $n \geq 1$, so $a_n \not\to 0$ $\Rightarrow$ divergent.

3. $\sum_{n=1}^{\infty} \boxed{\frac{4^n n!n!}{(2n)!}}$ $\qquad \frac{a_{n+1}}{a_n} \longrightarrow 1$

a_n $\qquad \Rightarrow$ The ratio test is inconclusive here.

But observe that $\qquad \frac{a_{n+1}}{a_n} = \frac{2n+2}{2n+1} > 1$ for all $n \geq 1$

$\Rightarrow a_{n+1} > a_n \qquad \Rightarrow a_n \not\to 0$ & so by the nth term test, the series diverges.

4. $\sum_{n=1}^{\infty} \frac{n^2(n+2)!}{n!3^{2n}} = \sum_{n=1}^{\infty} \frac{n^2(n+2)(n+1)}{3^{2n}}$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2(n+3)(n+2) \,/\, 3^{2n+2}}{n^2(n+2)(n+1) \,/\, 3^{2n}} \longrightarrow \frac{1}{9} \quad (<1)$$

$\Rightarrow$ So the series converges by the ratio test.

The Root Test

Consider the series with the terms

$$a_n = n/2^n, \; n \text{ odd}$$

$$a_n = 1/2^n, \; n \text{ even}.$$

What can ratio test tell us about $\sum a_n$?

$$\frac{a_{n+1}}{a_n} = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ even} \quad \to \infty \text{ as } n \to \infty \\ \frac{1}{2n} & \text{if } n \text{ is odd} \quad \to 0 \text{ as } n \to \infty \end{cases}$$

$\lim_{n\to\infty} \frac{a_{n+1}}{a_n}$ does not exist.

Observe that $\frac{n}{2^n} < \frac{1}{(1.5)^n}$ for n sufficiently large;

so we can actually use DCT here to conclude that $\sum a_n$ converges.

Theorem (The Root Test):

Let $\sum a_n$ be a series with $a_n \geq 0$ for $n \geq N$, and suppose that

$$\lim_{n\to\infty} \sqrt[n]{a_n} = \rho.$$

Then

(a). the series converges if $\rho < 1$,

(b). the series diverges if $\rho > 1$ or ρ is infinite,

(c). the test is inconclusive if $\rho = 1$.

For the previous example, for n even $\sqrt[n]{a_n} = \sqrt[n]{1/2^n} = \frac{1}{2}$

for n odd $\sqrt[n]{a_n} = \sqrt[n]{n/2^n} = \frac{n^{1/n}}{2}$

$\Rightarrow \quad \frac{1}{2} \leq \sqrt[n]{a_n} \leq \frac{n^{1/n}}{2}$ (Recall that $\lim_{n\to\infty} n^{1/n} = 1$)

$\Rightarrow \quad \lim_{n\to\infty} \sqrt[n]{a_n} = \frac{1}{2} \quad (< 1)$

So by the root test, the series $\sum_{n=1}^{\infty} a_n$ converges.

Proof of the Root Test: $\lim_{n\to\infty} \sqrt[n]{a_n} = \rho$.

(a) $\rho < 1$: pick $\varepsilon > 0$ such that $\rho + \varepsilon < 1$, then there exists N s.t. for all $n \geq N$, we have

$$|\sqrt[n]{a_n} - \rho| < \varepsilon.$$

$$\Rightarrow \sqrt[n]{a_n} < \rho + \varepsilon$$

$$\Rightarrow a_n < (\rho+\varepsilon)^n$$

If we define $c_n = (\rho+\varepsilon)^n$ for all $n \geq N$, then the geometric series $\sum_{n=N}^{\infty} c_n$ converges because $\rho + \varepsilon < 1$.

So by the DCT, $\sum_{n=N}^{\infty} a_n$ converges. $\Rightarrow \sum_{n=1}^{N} a_n$ converges

(b) $\rho > 1$: There exists N st $\sqrt[n]{a_n} > 1$ for all $n \geq N$

$\Rightarrow a_n \geq 1$ $\Rightarrow$ fails the nth term test

$\Rightarrow \sum a_n$ diverges.

(c) $\rho = 1$: For example,

- $\sum \frac{1}{n}$: $\sqrt[n]{\frac{1}{n}} = 1$ (diverges)
- $\sum \frac{1}{n^2}$: $\sqrt[n]{\frac{1}{n^2}} = 1$ (converges).

So when $\sqrt[n]{a_n} \to 1$, root test is inconclusive.

Examples:

1. $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$: $\sqrt[n]{a_n} = \frac{n^{2/n}}{2} \longrightarrow \frac{1}{2} \; (< 1)$.

Root test $\Rightarrow$ converges.

2. $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$: $\sqrt[n]{a_n} = \frac{2}{(n^{1/n})^3} \longrightarrow 2 \quad (> 1)$.

Root test $\Rightarrow$ diverges.

3. $\sum_{n=1}^{\infty} \left(\frac{1}{1+n}\right)^n$: $\sqrt[n]{a_n} = \frac{1}{n+1} \to 0 \quad (<1)$

Root test $\Rightarrow$ converges.

4. $\sum_{n=1}^{\infty} \underbrace{\left(1 - \frac{1}{n}\right)^{n^2}}_{a_n}$

Root test: $\sqrt[n]{a_n} = \left(1 - \frac{1}{n}\right)^n$

Recall that

$\left(1 + \frac{k}{n}\right)^n \to e^k$

$\to e^{-1} \quad (< 1)$

So Root test $\Rightarrow \sum a_n$ converges.

Extra Examples:

1. $\sum_{n=1}^{\infty} \frac{n^5}{5^n}$ converges

⑥ $\sum_{n=1}^{\infty} \left(\frac{4n+3}{3n-2}\right)^n$ diverges.

2. $\sum_{n=1}^{\infty} \frac{n^n}{2^{n^2}}$ converges.

3. $\sum_{n=2}^{\infty} \frac{n}{(\ln n)^n}$ converges.

4. $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges.

$k^n \ll n^n$

$\Downarrow\Downarrow$

$n! \sim \sqrt{2\pi n}\, n^n e^{-n}$

5. $\sum \frac{(2n)!}{(n!)^2 5^n}$ converges.

Ratio Test

$\sum_{a_n \neq 0} a_n$, $\lim \left|\frac{a_{n+1}}{a_n}\right| = \rho$

- $\rho < 1 \Rightarrow \sum a_n$ converges

- $\rho > 1$ or $\rho \to \infty \Rightarrow \sum a_n$ diverges

$|a_{n+1}| \geq |a_n|$

Given a series $\sum_{n=1}^{\infty} a_n$.

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum a_n$ converges.

$\frac{|a_{n+1}|}{|a_n|} \to \rho < 1 \Rightarrow$ convergence

WARNING : $\sum a_n$ converges $\not\Rightarrow$ $\sum |a_n|$ converges

$$a_n = (-1)^{n+1} \frac{1}{n}$$

$$\sum a_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

(later) $\sum a_n$ converges.

but $\sum |a_n| = \sum \frac{1}{n}$ diverges.